Engineered Intelligence: Tracing Computational Design from Search to Evidence

A Retrospective Study of Representation, Search, and Assessment

Research draft prepared for Jepson Taylor | Human author approval pending

Abstract—Automated design searches can retain extensive candidate records while leaving the evidence behind a selected result difficult to interpret. We present Jarambit as a retrospective study linking an application-specific geometric representation, recorded search observations, and candidate-specific numerical evidence. The largest archive contains 1.95 million search-evaluation records; historical genealogy and intermediate screening are analyzed separately because their scoring and retention regimes differ. Three findings delimit what these records establish. Evidence coverage is uneven: extensive genome and score tables do not imply retained parentage or downstream assessments. Recorded historical ancestry exposes ancestor reuse, while the highest stored search score belongs to a candidate that fails a later internal assessment. For a separately assessed export, selected solid-model observables agree closely between implementations on a shared mesh, while separate plate-refinement sequences do not admit the chosen extrapolated-error estimates. The contribution is an inspectable connection between representation, candidate identity, assessment stage, and source artifacts, with incompatible scores and missing observations kept explicit. These distinctions make the reported evidence easier to question and reconstruct. The findings remain computational and retrospective; physical performance and general search advantage are unestablished.

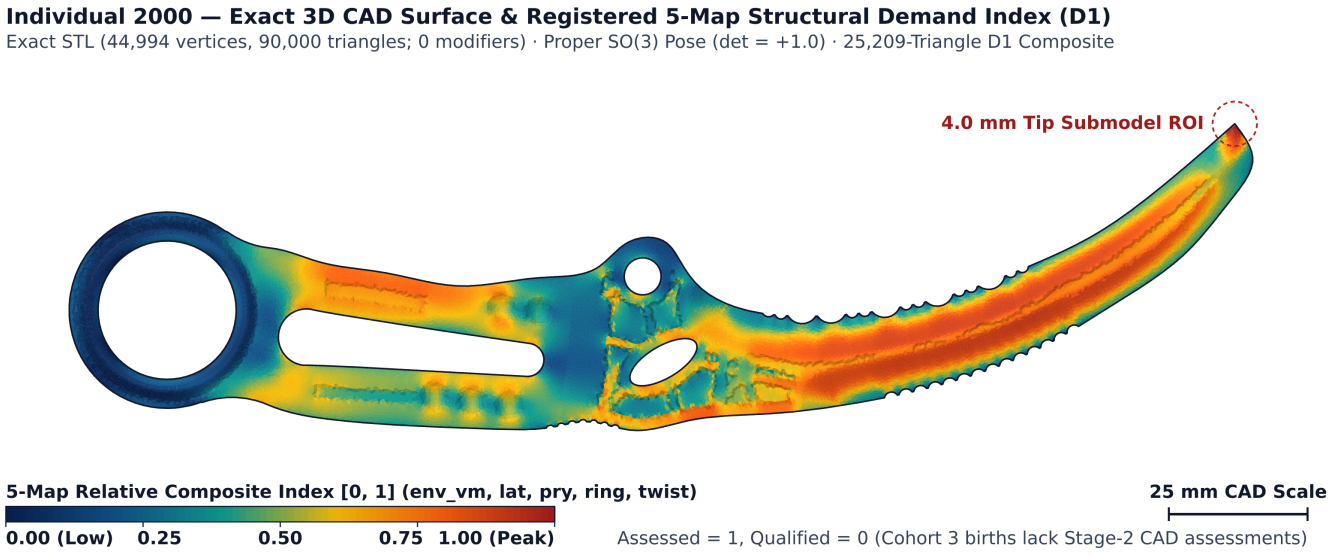


Fig. 1. Individual 2000, the illustrated assessed export. Colors show a candidate-specific, dimensionless composite of retained response fields; high values do not denote safety or qualification. Contributor availability varies across the surface. Definitions and support accounting appear in Sections VI and IX.

Index Terms—Engineered Intelligence, computational design, evolutionary computation, executable genomes, CAD, quality diversity, finite elements, evidence provenance, reproducible populations.

Research draft; author approval pending. Retained computational evidence, with physical observations not reported in this study. Source and supplement accessions are described in Section X.

I. Introduction: Tracing Search to Evidence

AUTOMATED design searches can produce more candidate records than a reader can inspect individually. A selected geometry and its final score reveal little about which representation produced it, which objective ranked it, or whether later assessments support the same conclusion. Interpreting a computational design result therefore requires links between the candidate, its search context, and the evidence actually retained for it.

We use *Engineered Intelligence* for a design process connecting a versioned representation, a geometric decoder, recorded variation and selection, declared objectives, and inspectable observations. Jarambit provides a retrospective case study of these connections in an application-specific titanium CAD system. Its representation has 128 numeric coordinates, including discretized selectors, plus a categorical material field. The analyzed records come from workflows with different scorers and retention policies; they do not form a single uniform experiment.

A. Three Questions and Findings

This study asks three questions about what the retained records establish:

1) What can a reader reconstruct about a generated individual and the search context that ranked it?
2) Which evidence is retained for each cohort, and what is observed when an individual reaches later assessment?
3) What do selected numerical agreement and mesh-study observations establish, and what remains unresolved?

Three findings answer these questions. First, evidence coverage is uneven: the large native-search archive retains compact genome and score records, while recorded parentage belongs to a smaller historical cohort. Neither collection's size establishes downstream assessment coverage. Second, the historical genealogy exposes ancestor reuse, and its highest stored search score belongs to a birth that fails a later internal assessment. This is a documented assessment disagreement; the retained example does not determine a general failure rate or a pairwise rank reversal. Third, selected solid-model observables agree closely between two implementations on a shared mesh, while separate plate-refinement sequences do not admit the chosen extrapolated-error estimates. These observations answer different numerical questions.

The historical genealogy, intermediate screening campaign, completed native runs, and separately assessed exports are therefore reported with their own identities and evidence coverage. In particular, the illustrated assessed export, Individual 2000, supplies the spatial fields; Individual 2002 supplies a retained weighted reporting-profile record whose exported CAD is unavailable. Their observations are not interchangeable. Section VII develops the cohort-specific findings, and Section IX interprets the numerical evidence.

B. Three Bounded Contributions

1) **Representation and identity links.** A semantic genome contract and distinct identities for births, memberships, and assessed artifacts let readers ask how a recorded individual was produced. The historical ancestry example demonstrates reuse across branches rather than an assumed branching tree (Sections IV–V).
2) **Retrospective interpretation across assessment stages.** Keeping score regimes and missing observations separate lets readers ask which conclusions a retained ranking supports. The historical search-to-assessment disagreement and explicit unscreened proposals demonstrate why a preferred, unobserved, or assessed candidate denotes a different evidentiary state (Sections VI and VII).
3) **Source-bound evidence views and coverage accounting.** Linked records, field views, and reporting artifacts let readers inspect the source and scope of a displayed result. The shared-mesh comparison and refused plate extrapolations demonstrate how distinct evidence can be presented without turning it into a single validation claim (Sections IX–X).

Study scope. This is a retrospective study of retained computational evidence. Internal assessment labels describe the recorded workflow; physical observations are not reported in this study. The records do not establish comparative optimization superiority, physical qualification, or transfer to other design domains. Reporting reconstruction and numerical solver replay remain distinct operations.

II. Background and Related Research

A. Computational Design and Representation

Computational design already includes mature approaches to changing structural form. Sigmund and Maute review density, level-set, evolutionary, and other topology-optimization approaches [13]. Jarambit's documented native schema instead varies an application-specific parameterization under a fixed decoder. Its coordinate count is not evidence of an unrestricted topology search or a newly discovered representation language.

DeepCAD learns CAD-operation sequences for autoencoding and generation [2]; Text2CAD generates parametric CAD sequences from text and reports visual, parametric, and geometric evaluations [3]. These representation choices differ from Jarambit's numeric schema. None supplies a common mechanical benchmark for the present study.

Evolutionary engineering also predates language-model assistance. The cited 2003 ST5 report describes evolved antenna representations, fabricated prototypes, and comparison with a hand-designed antenna; flight qualification was still pending in that report [4]. MAP-Elites retains representatives across user-defined behavioral dimensions [5]. Such archive selection differs from retaining all observed births, and archive occupancy alone does not measure population diversity. Jarambit's cohort-specific archive and score definitions therefore remain part of its reported context.

B. Research Provenance and Artifact Access

The closest context for this study's evidence links is research provenance. PROV-DM distinguishes entities, activities, derivations, responsible agents, and collections [14]. These concepts

help explain why a candidate record, a search activity, an exported geometry, and a reporting derivative require distinct identities. Jarambit's retained links are compared on these dimensions; no PROV conformance or general interchange implementation is established here.

The FAIR principles address discovery, access, interoperability, and reuse of research objects [15]. Software citation principles additionally distinguish persistent identification, access, and specificity of the software version used [16]. A hash identifies bytes; meaningful reconstruction also requires retrievable artifacts and interpretation of their types, revisions, and relationships. This study's contribution is a concrete retrospective account of those relationships and gaps, rather than a new general provenance model.

We cite the inspected Jarambit publication baseline as a source snapshot [1]. It is distinct from the cohort producers and from the final manuscript release accession (Section X). A methods publication, a source snapshot, an executed binary, and a generated figure likewise denote different objects.

C. Numerical Credibility and Visual Interpretation

NASA's models-and-simulations standard supplies context for separating credibility questions; compliance is not claimed [7]. The scikit-fem and DOLFINx methods references identify finite-element software roles, while execution provenance must identify the versions and model inputs actually used [8], [9]. Selected observable agreement on a shared mesh and admissibility of a grid-refinement estimate answer different questions (Section IX).

Quantitative views also need explicit units, transformations, masks, and legends. Crameri *et al.* explain how uneven color gradients can introduce visual distinctions absent from the data [17]. A percentile summary therefore requires its own interpretation and cannot inherit the physical units of its component fields. The paper's spatial views are reporting devices, not a separately validated visualization method.

III. Study Design and Retained Evidence

This is a retrospective study of selected retained records, rather than a prospective comparison of optimizers. We ask what those records establish about representation, recorded search preference and later numerical assessment. The three cohorts have different scoring and retention policies; the separately assessed exports supply additional candidate-specific observations, rather than downstream measurements for every search birth.

A *birth* is a recorded candidate-creation event. A *membership* is an appearance in an island slot, including retention or migration of an existing individual. An *archive representative* is a selected record under its archive's objective and descriptor context. An *assessed export* has a specifically bound artifact/assessment bundle. These are different counting units: one genome can occur in several births or memberships, and a certificate identity alone does not establish that its CAD or assessment artifact is retained.

Sources H and P are the historical reconciliation and proposal/screen join accessions; N is the completed generation-800 population accession. E is the automatic manifest plus original field/assessment/verification contracts; C is the controlled-export scorecard accession. Availability is local to these accessions. The native population tables retain all 128 numeric coordinates and material, but their size does not establish complete parentage, per-birth exported CAD, later solid assessment or independent physical replication.

A. Run Identities and Capture Overlap

The native inventory contains 18 current campaign directories: ten completed generation-800 runs and eight others. An earlier staged snapshot has eleven summary files, of which four are earlier captures of runs later included among the ten completed runs (public Runs 6–9). These are overlapping views of original run identities, not 21 independent runs. Nine aborted crescent attempts are a separate accession; seven retain 216 summary rows in total and two have no retained CSV rows. Summary rows describe generation/island aggregates, not new births. Capture/receipt times are kept separate from unverified historical process start/stop times.

B. Selection and Interpretation

Individual 2000 was selected editorially because its retained bundle connects export geometry, plate fields, mesh-study observations, solid comparison and scorecard. This selection does not confer championship or qualification. In the historical cohort, the highest stored birth score identifies an illustrative later-assessment disagreement. Neither selection is a random sample from the large native archive. Missing observations remain missing; they are not failures, favorable outcomes or a measured qualification fraction.

IV. Representation and Candidate Identity

A. A Gene Is a Semantic Contract

A gene requires an explicit coordinate index, physical or discrete type, unit, domain, coupling law, and decoder revision. A serialized floating-point token alone cannot distinguish a millimeter displacement from an angle, a feature-count selector, or an architecture switch. Those semantics determine what a stored genome means when it is decoded.

The documented native schema stores 128 numeric coordinates and one categorical material field. Some numeric coordinates express continuous parameters; others are interpreted through thresholds or discretization. The nominal coordinate count is therefore neither the number of independently active parameters nor a measure of geometric freedom. Field roles are summarized below; exact domains and decoding rules require the producer revision attached to each cohort. Historical-producer details are retained in Supplement I, without attributing them to every executed workflow.

B. Coupling and Conditional Activity

Clamping, thresholding, and coupled bounds distinguish stored coordinates from the geometry they produce. In the bound historical producer, the continuous-rail switch is thresholded, the active knot count is floored and bounded, and

TABLE I
CONTEXT BY REPRESENTATION, SEARCH, EVIDENCE, AND VALIDATION SCOPE. ROWS DESCRIBE DISTINCT RESEARCH PURPOSES RATHER THAN A PERFORMANCE RANKING.

Context	Representation	Search or transformation	Evidence emphasized	Validation scope
Topology optimization	Varies by formulation	Numerical design updates	Structural objectives and constraints	Method- and model-specific
DeepCAD / Text2CAD	CAD-operation sequences	Learned generation	Reconstruction or generated CAD	Reported geometric/parametric evaluations
PROV / FAIR	Research objects and relations	Derivation and stewardship	Identity, relationships, and access	Provenance/access context, not mechanics
This study	Fixed CAD schema and candidate identities	Retained evolutionary and assessment records	Coverage, ancestry, scores, and selected numerical observations	Retrospective computational evidence

TABLE II
COVERAGE OF THE ADMITTED STUDY RECORDS. A: AVAILABLE; P: PARTIAL/SELECTED; NR: NOT RETAINED IN THE NAMED ACCESSION; U: COVERAGE NOT ESTABLISHED. PHYSICAL OBSERVATIONS ARE NOT REPORTED IN THIS STUDY (NPUB); THIS DOES NOT INFER THAT NO EXPERIMENT EVER OCCURRED. EACH ROW'S SOURCE KEY APPLIES TO ALL ITS CELLS; THE SUPPLEMENT SUPPLIES EXACT PATHS, HASHES AND QUALIFICATIONS.

Records (source)	**Denominator**	**Genes/ material**	**Parentage**	**Exported geometry**	**Spatial fields**	**Later assessment**	**Physical**
Historical genealogy (H)	1984 births; 2176 memberships	A	A	P	P	P: selected strict records	NPub
Proposal/screen campaign (P)	3649 proposals; 2259 screens	A in proposals	P	P: selected exports	P	P; not every proposal	NPub
Ten completed native runs (N)	1,954,560 births	A	U beyond admitted tables	P: separate STL accession	NR per birth	NR per birth	NPub
Automatic assessed export (E)	Individual 2000; one assessed	A	U birth link	A	A plate; NR full solid	A saved bundle	NPub
Controlled exports (C)	Individuals 2001/2007, separately scoped	A	U birth link	A	P	A saved scorecards	NPub

TABLE III
FUNCTIONAL GROUPS IN THE DOCUMENTED NUMERIC SCHEMA.

Group	Role and stored type	Conditional interpretation
Silhouette	Numeric normal offsets around a reference outline	Some stations can be pinned by the producer's bounds.
Planform and features	Dimensions, angles, ratios, and feature selectors	A stored numeric feature setting need not activate a feature.
Internal machining	Pocket, window, and perforation parameters	Feature counts and zone switches use discrete decoding rules.
Continuous-rail layer	Numeric architecture switch, dimensions, and thickness knots	The switch selects the thickness architecture; only the selected knot count contributes.

enabling the layer disables several legacy features. When it is disabled, knot coordinates are set to stock thickness during clamping. The decoder and its configuration thus belong to the interpretation of a record, rather than being incidental implementation details. A numeric change can be inactive, discretized away, or constrained by another coordinate; changed-coordinate counts alone do not establish geometric change.

C. Birth, Genome, Membership, and Export

A birth identifies a recorded generation event; a genome serialization identifies stored coordinate and material values; a membership identifies a population-slot occurrence; and an assessed export identifies its own geometry and assessment bundle. These objects can share an individual identifier without sharing a denominator. The historical cohort, for example, retains repeated genome serializations and repeated appearances of earlier births. A display mesh, a plate mesh, and a solid submodel likewise represent distinct discretizations, each requiring its own source binding (Section X).

The inheritance ledger counts changed numeric coordinates by exact numeric inequality on parsed CSV values. Its raw-coordinate L_2 norms mix units and discrete semantics. They are audit summaries of stored coordinates, not physical distances between shapes; the detailed edge ledger is retained in Supplement I.

V. RECORDED SEARCH AND INHERITANCE

A. Four Collections with Different Meanings

Births, population memberships, archive representatives, and assessed exports answer different questions. Births record proposed individuals and their observations. Memberships record occupied island slots, including retained individuals. Quality-diversity archives retain representatives under a run's descriptor and score rules. Later assessments concern selectively admitted or separately exported candidates. Evidence coverage and denominators for these collections are reported with the cohort-specific results (Section VII).

For the historical cohort, the recorded membership stage is *after contextual rescoring and before ring migration*. The bound producer emits birth or elite events and summary observations before migration. Migration subsequently copies selected IDs to destination islands; the overwritten destination IDs are not retained by those events. Calling the earlier snapshot a generation-end post-migration population would therefore overstate what can be reconstructed. Elite events retain membership but omit refreshed contextual fitness, so an individual's stored birth score is not substituted for an unrecorded later score.

B. Variation in the Historical Producer

The historical producer uses four island populations, shared novelty memory, a quality-diversity archive, elite retention, tournament selection, and periodic ring migration. Parents can be selected from the archive or current population; recorded parent IDs and operation labels identify the observed birth mechanism. These are source-bound historical semantics, rather than an assertion that every retained cohort executed an identical binary.

Crossover inherits the silhouette's 40 control points as one block and chooses the remaining numeric coordinates and material category independently between parents, then clamps the result. Mutation selects among six modes: sparse coordinate perturbation, a smooth silhouette bump, a thickness-and-section operator, a lattice operator, a fuller/texture/feature operator, and a machining-or-rail operator. Mutation scale depends on island and configuration; the exact parameter branches and the coarse/fine distinction are retained in Supplement I. This is variation within a fixed representation and decoder.

C. Ancestor Reuse in the Retained Ledger

Figure 2 uses the current generated inheritance ledger's three-hop view. Individual 1095 has recorded parents 413 and 526, and 526 itself has parents 413 and 392. Individual 413 is therefore reused along the paths $413 \to 1095$ and $413 \to 526 \to 1095$. The dashed $174 \to 299$ link is a recorded relationship retained as induced closure, distinct from the three-hop edges. The ledger preserves exact coordinate differences and parent ordinals; the main figure emphasizes relationships rather than repeating their raw mixed-unit magnitudes.

This ancestry illustrates a reconstruction capability. Its focal birth's high stored search score and later assessment disagreement are separate observations, developed in Section VII; neither ancestry nor score alone establishes physical performance.

VI. Score Definitions and Comparability

A search preference, a stored reporting vector, a shortfall score and a mechanical safety factor answer different questions. Their names, directions, observation stages and missing-input rules must accompany their values. Table IV keeps the score regimes separate; a shared word such as "fitness" does not create a common performance axis.

A. A Retained Weighted Reporting Vector

The principal reporting definition is

$$F = \sum_{i=1}^{20} w_i s_i, \qquad s_i \in [0,1], \quad \sum_i w_i = 100. \tag{1}$$

Its priorities are mass/compactness (30%), form/comfort proxies (25%), exported geometry (15%) and response/verification (30%). Table V shows sums of *stored weighted contributions* for Individual 2002. The Supplement retains all twenty normalized values, weights and contributions. These are reported observations from its summary, not a recalculation under the present scorer and not measurements transferred from Individual 2000.

The saved scalar is $F = 74.207299$, with reported raster mass $101.139\,\mathrm{g}$ and solid-mass scalar $101.1384\,\mathrm{g}$. Several terms are clipped ratios, Boolean values or proxies. In particular, a stored solid or support term in this reporting vector does not establish a retained underlying CAD/solid assessment artifact. The retained directories lack Individual 2002's exported geometry and round populations; its saved qualification label is "pending independent top-k assessment." Individual 2000 instead supplies the illustrated assessed geometry and response fields.

For the assessed Individual 2000, the actual failed-branch score is $W_{\mathrm{CAD}} = -5.570408$. The same scorecard stores unpenalized quality 1.456983; that is the hypothetical pass-branch value, not an additional achieved score. The Supplement gives the branch equations, input names and shortfall arithmetic. Those scores and G_1 remain separate from F and native f.

B. Context and Display Direction

Native fitness includes novelty computed against contemporaneous population/archive descriptors. A value can therefore change when its context changes, even if the genome does not. Birth scores and later membership observations also have distinct stages. Cross-generation or cross-run maxima describe retained contextual preference; they do not isolate a physical improvement. The historical and newer native implementations/configurations require separate provenance even when both call their scalar f.

A display loss $L = -f$, $100 - F$, or $-W_{\mathrm{CAD}}$ reverses numerical order. Minimizing it preserves the original preference for a larger native score. It does not make those different profiles comparable. A cumulative maximum is monotone by construction, so its corresponding loss is nonincreasing by construction; that fact alone is not evidence of continuing physical improvement.

C. Relative Field Summary and Physical-Unit Views

The hero summarizes retained fields of Individual 2000. For each active map, cells are sorted by stored magnitude and assigned the midpoint of their cumulative area interval. Equal stored values are processed in deterministic source-cell order, rather than pooled into one tied percentile. The active-map ranks are averaged at each cell, then the candidate's resulting minimum and maximum are linearly stretched to $[0,1]$ for display. This candidate-specific relative summary removes

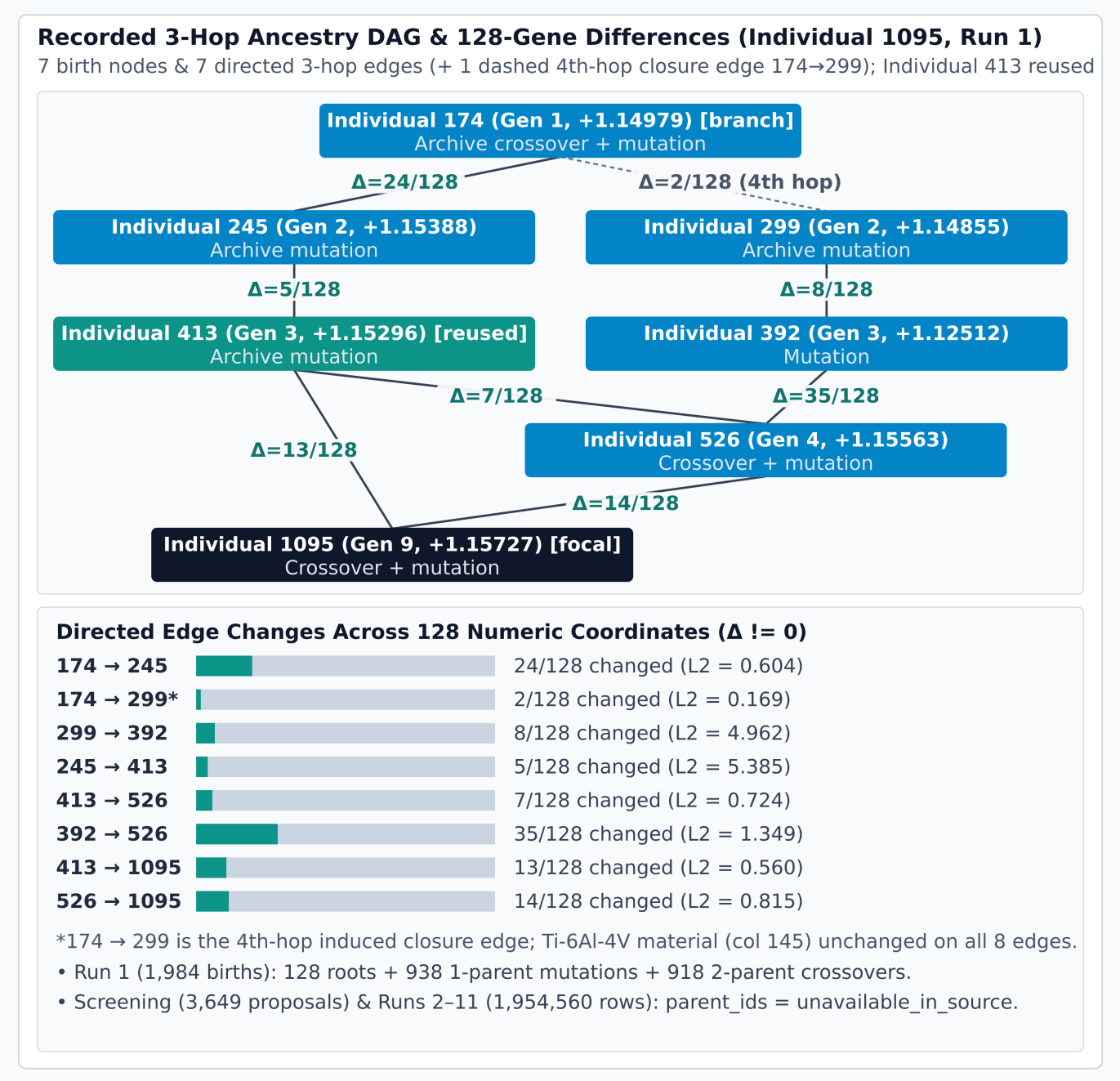


Fig. 2. Shared ancestry in the historical search record around Individual 1095. Solid edges show recorded parent-to-child links within three parent hops; the dashed edge 174 → 299 closes the induced subgraph but lies four hops from the focal individual along that path. Individual 413 contributes through both a direct and an indirect path. The genealogy belongs to this historical cohort.

physical units. A color value of 0.8 means neither 80% safety nor an absolute 80th percentile shared by other candidates.

The five maps are the envelope, lateral, pry, ring and torsion arrays. Because the envelope is derived from the primitive responses, five contributors do not represent five independent tests. Ring/envelope masks vary the number of active contributors; the cover does not display a separate per-cell support map. Section IX and the Supplement describe the actual support counts and primitive-only sensitivity.

Figure 3 retains four physical-unit maps; the fifth, in-plane tip pry, appears in Figure 4C. Gray denotes producer-masked cells, not measured zero stress. Raw torsion has a large tip-patch peak, which remains distinct from the post-exclusion peak used by selected capacity calculations.

VII. Three Findings from the Retained Records

A. Large Record Coverage Does Not Imply Rich Assessment Coverage

The ten completed native runs (Runs 2–11) retain 1,954,560 evaluated birth records, 195,456 per run across generations 0–800. The admission receipts separately count 32,040 generation/island summary rows. All birth rows preserve the

TABLE IV
SCORE DICTIONARY. SCORES ARE DIMENSIONLESS; PHYSICAL UNITS REMAIN ATTACHED TO THEIR INPUTS. MISSING OBSERVATIONS DO NOT BECOME ZERO SCORES.

Quantity / stage	Direction / scale	Meaning and normalization	Comparison / missingness
Native f; historical and completed native search	Higher preferred; conditional negative branches, no common $[0, 1]$ scale	Mechanical/mass terms and contextual novelty after legality/feasibility branches	Bound profile, configuration and observation stage required; novelty changes with archive context; sign is not legality
Screening deficit; intermediate proposals	Lower shortfall preferred; nonnegative stored sum	Stage-1 screening quantity; not the later 20-term reporting vector	Only observed screens have numeric values; 1390 proposals lack screens
Weighted F; retained reporting curves	Higher preferred; $[0, 100]$	Twenty saved normalized terms with weights totaling 100	Original producer/scorer revision and point-stage comparability not established by summaries; no invented CAD or population distributions
W_{CAD}; 16-gate CAD scorecard	Higher preferred; conditional positive/negative branches	Pass-branch quality versus failed-branch normalized-shortfall penalty	Same assessment definition/inputs required; a favorable subscore is not all-gate success
G_1; assessment composite	Higher preferred; dimensionless	Weighted solid-tip factor, strength/mass, shape, mass penalty and sequence gap	Missing required inputs produce incomplete status; a screen-tip substitute is marked provisional; does not override gates

TABLE V
INDIVIDUAL 2002: RETAINED REPORTING PRIORITIES AND CONTRIBUTED POINTS. WEIGHTS EXPRESS RESEARCH PREFERENCES; THEY ARE NOT PROBABILITIES OF SAFETY OR QUALIFICATION.

Objective group	Weight (%)	Points
Mass and compactness	30	15.470
Form and comfort proxies	25	23.875
Exported geometry	15	11.841
Response and verification	30	23.022
Total	100	74.207

numeric coordinates and material, but no per-birth later CAD/solid assessment is retained for this population. The largest archive therefore supplies search-record coverage rather than a qualification rate.

Figure 5 keeps the native population separate from the historical 1984 births and the intermediate campaign's 2259 numeric screens. The 1390 proposals lacking screens have no invented score. Within the native archive, 754,236 births (38.59%) are search-legal; 610,051 of those have negative contextual fitness. Score sign consequently cannot stand in for legality. Per-run legal counts range from 70,278 to 80,132, with mean 75,424 and sample standard deviation 2937 births. Run-maximum stored fitness ranges from 1.1089 to 1.1440 (mean 1.1250, sample standard deviation 0.01285). These are descriptive summaries of ten selected runs; related births do not provide millions of independent replicates.

B. Recorded Ancestry Enables a Later-Assessment Disagreement to Be Inspected

The historical Run 1 retains 1984 births, complete recorded parent links, 2176 island-slot memberships and 12 migration events. The genealogy exposes reuse of ancestors rather than a simple branching tree. Its saved FEM-status categories distinguish 1455 complete records from 529 not evaluated; the latter are not failed solves.

Individual 1095 has the highest stored historical birth score ($f = 1.15727$, generation 9). Its later strict-assessment summary records cert_pass_all=false and seq_ok=false, although its strict lateral predicate is true. This is a documented disagreement between stored search preference and a later assessment criterion. It does not demonstrate a pair of candidates reversing rank or establish a general rank-transfer failure rate.

The intermediate campaign retains 3649 proposals in 57 birth files across 24 run directories. One-based candidate indexing joins 2259 screens; 1390 screen observations are missing. A separate retained weighted-reporting subset has six curve files and fifteen generation summaries. In Run 31, the archive score changes from 74.1675 (Individual 2009) to 74.2073 (Individual 2002), then stays at that value. An archive maximum is monotone by construction. The curve alone does not show continuing improvement in physical quality, and its unestablished producer/stage comparability precludes treating it as a controlled convergence experiment.

The automatic leaderboard separately retains one assessed export, Individual 2000, with zero qualified and no promoted champion. The historical 23×18 battery includes shared controls, blocked bindings and completeness flags; it is not a set of 414 independent mechanical tests. Individual 2002 is a reporting record with absent retained CAD, rather than an additional assessed export. Its vector is not transferred to the geometry or response fields of Individual 2000.

C. Selected Solver Agreement and Refused Plate Extrapolation Support Different Conclusions

On the same saved solid mesh, the scikit-fem/PyAMG and DOLFINx/PETSc paths agree within 0.05% for the two retained recovered safety-factor observables. Separately, the plate mesh sequence does not admit its selected extrapolated error estimates: one observed order is nonpositive and the other sequence oscillates. Section IX gives the inputs, recovery definitions and refusal reasons. Close shared-input agreement



Fig. 3. Spatial concentration differs across Individual 2000's retained plate responses. A: scaled multi-case envelope (MPa). B: unit lateral response (MPa/N). C: unit ring traction (MPa/N). D: raw unit chord torsion (MPa/(N mm)). The plate has 25,209 triangles; gray marks the envelope's 75 tip-excluded cells and ring field's 14,875 seated cells. Tip pry, the fifth composite component, is shown separately in Figure 4C. Each field keeps its own physical scale.

TABLE VI
CANDIDATE ROLES AND RETAINED EVIDENCE. A PUBLIC IDENTIFIER OR REGISTRY CATEGORY DOES NOT SUBSTITUTE FOR AN ACTUAL RETAINED GEOMETRY OR ASSESSMENT ARTIFACT.

Individual	Role	Retained evidence	Interpretation
1095	Highest stored Run 1 birth score	Birth, parentage and later strict-summary record	Later assessment not met; no pairwise rank reversal
2000	Editorially illustrated automatic export	Bound CAD, plate fields, numerical observations and assessment bundle	One assessed export; not qualified
2001 / 2007	Separately assessed controlled exports	CAD/scorecard records; $W_{\mathrm{CAD}} = -3.879/-3.965$ and $G_1 = 0.624/0.622$	Grouped strict verification not met
2002 / 2009	Weighted reporting records in Run 31	Stored summary scores $74.207299/74.167488$	CAD and underlying round populations absent from retained reporting directories

does not repair the extrapolation assumptions or establish physical qualification.

VIII. WHAT THE NUMERICAL OBSERVATIONS REPRESENT

The retained mechanical observations answer different questions. A unit-load plate field, a later assessment scalar, a shared-mesh solver comparison, and a software-control flag

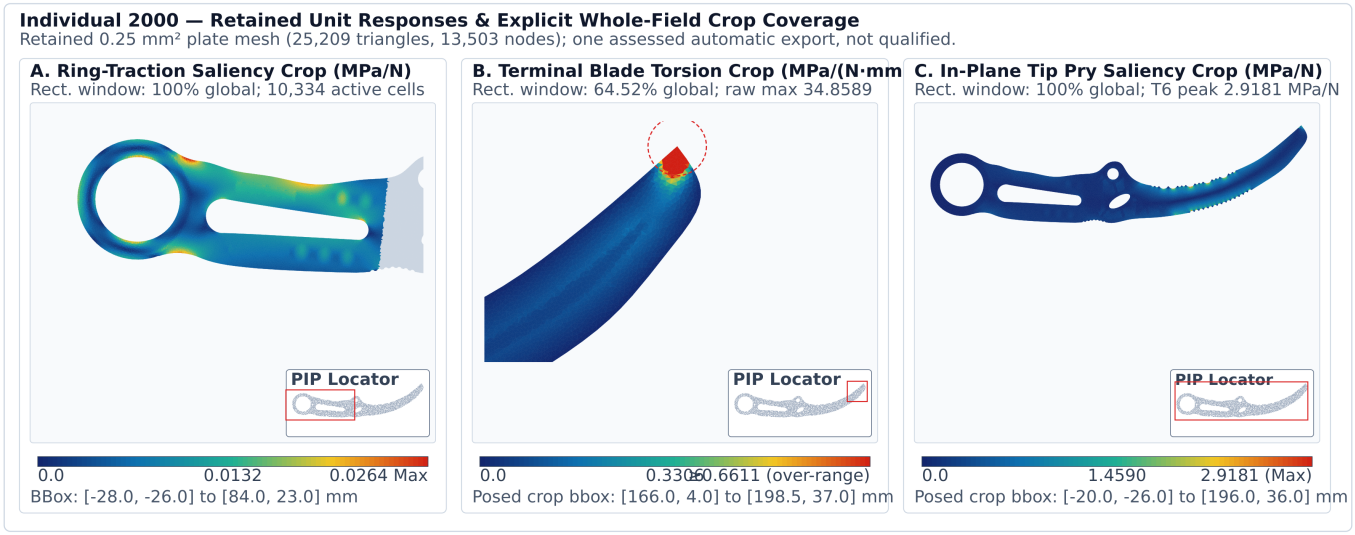


Fig. 4. Individual 2000 retained response views with whole-object locators. A: ring traction (MPa/N). B: terminal chord torsion (MPa/(N mm)); this is a restricted terminal view, without a global 90% coverage claim. C: in-plane tip-pry T6 centroid response (MPa/N), with stored peak $2.9181\,\mathrm{MPa/N}$. The raw torsion peak exceeds the post-exclusion capacity-scale range; its over-range treatment and full-range reference must be read together. Domains, masks and integration rules are distinguished in the Supplement; final achieved fractions require the bound emitted-domain receipt.

do not provide interchangeable evidence. Table VII groups the eleven observation families by the conclusion each can support. The detailed formulas, source keys, and historical catalog are retained in the Supplement; the fixed 23-subject historical battery is separate from the single automatically assessed export, Individual 2000.

A. Plate Fields and Assessment Stages

For Individual 2000, the retained plate mesh contains 13,503 nodes and 25,209 triangles. Lateral bending and chord torsion use DKT bending; in-plane tip pry and ring traction use quadratic T6 membrane elements. Their unit-response maps have distinct units and load definitions (Section IX). The scaled multi-case envelope is derived from those maps; it is not an independent fifth load case.

The assessment comparison in Figure 6C is *certification versus strict*, not a generic coarse/refined pair. Certification takes the minimum capacity over maximum triangle areas 0.25 and $0.125\,\mathrm{mm}^2$, while strict assessment uses $0.0625\,\mathrm{mm}^2$. The saved lateral capacities are 168.90 and $168.18\,\mathrm{N}$; the local tip-torsion factors are 0.524 and 0.529. The displayed fields come from the $0.25\,\mathrm{mm}^2$ level. These stages are distinct from the $4/1/0.25\,\mathrm{mm}^2$ plate mesh study.

The global envelope excludes cells within $4\,\mathrm{mm}$ of the tip. The plate body-torsion observable searches beyond $15\,\mathrm{mm}$ from the tip; the local tip observable searches the 4–$15\,\mathrm{mm}$ annulus. Raw torsion imagery retains the tip-patch response and must distinguish its raw peak from the post-exclusion capacity peak. Ring-traction cells in the producer's seated region are masked, rather than interpreted as unstressed material.

B. Local Solid Response and Recovery

The retained P_2 solid submodel uses $91{,}008$ tetrahedra and $416{,}583$ displacement degrees of freedom. Its cut-arc displacement degrees of freedom are clamped, and vertex forces in a $2\,\mathrm{mm}$ tip patch apply $1.82\,\mathrm{N\,m}$ chord torque. The retained comparison is scikit-fem/PyAMG CG versus DOLFINx/PETSc CG/GAMG on that same saved submodel. Rust orchestration is not the primary solid assembly implementation. Section IX specifies the common inputs, recovery and selected agreement; the publication inputs contain scalar and binned profiles, rather than a full three-dimensional stress field.

C. Proxies, Support Assumptions and Missing Observations

The regional yield-order ratio compares modeled elastic capacities and has a stored target of 1.2. It does not model plastic redistribution, cracking or a physical failure sequence. Likewise, the modified-Goodman fatigue factor ($n_f = 0.472$) is a scalar model result, not an observed cycle life.

An alternate support study uses bore-bearing springs and a Winkler grip-pad foundation. Its lateral regional ratio (5.203) and stored buckling eigenvalue (7.035) remain separate from the certification multi-case ratio (1.435). The differing support/load observables are not a mesh-convergence pair. The eigenvalue is provisional because the operator-symmetry/eigenpair audit remains open. Figure 6A presents support assumptions and scalar records; no support eigenvector field is retained.

D. Geometry Records and Control Catalogs

Export identities and reference-contour comparisons bind what was generated and measured. They establish neither physical performance nor human comfort. The historical 23×18 battery and 44-entry crosswalk remain available in the Supplement. They mix candidate-specific observations, shared software controls, missing bindings and completeness flags. In particular, its 20 quadratic-solid completeness labels do not denote 20 favorable strength margins. Catalog definitions, executed controls and unavailable records are reported separately; their counts are not a mechanical replication denominator.

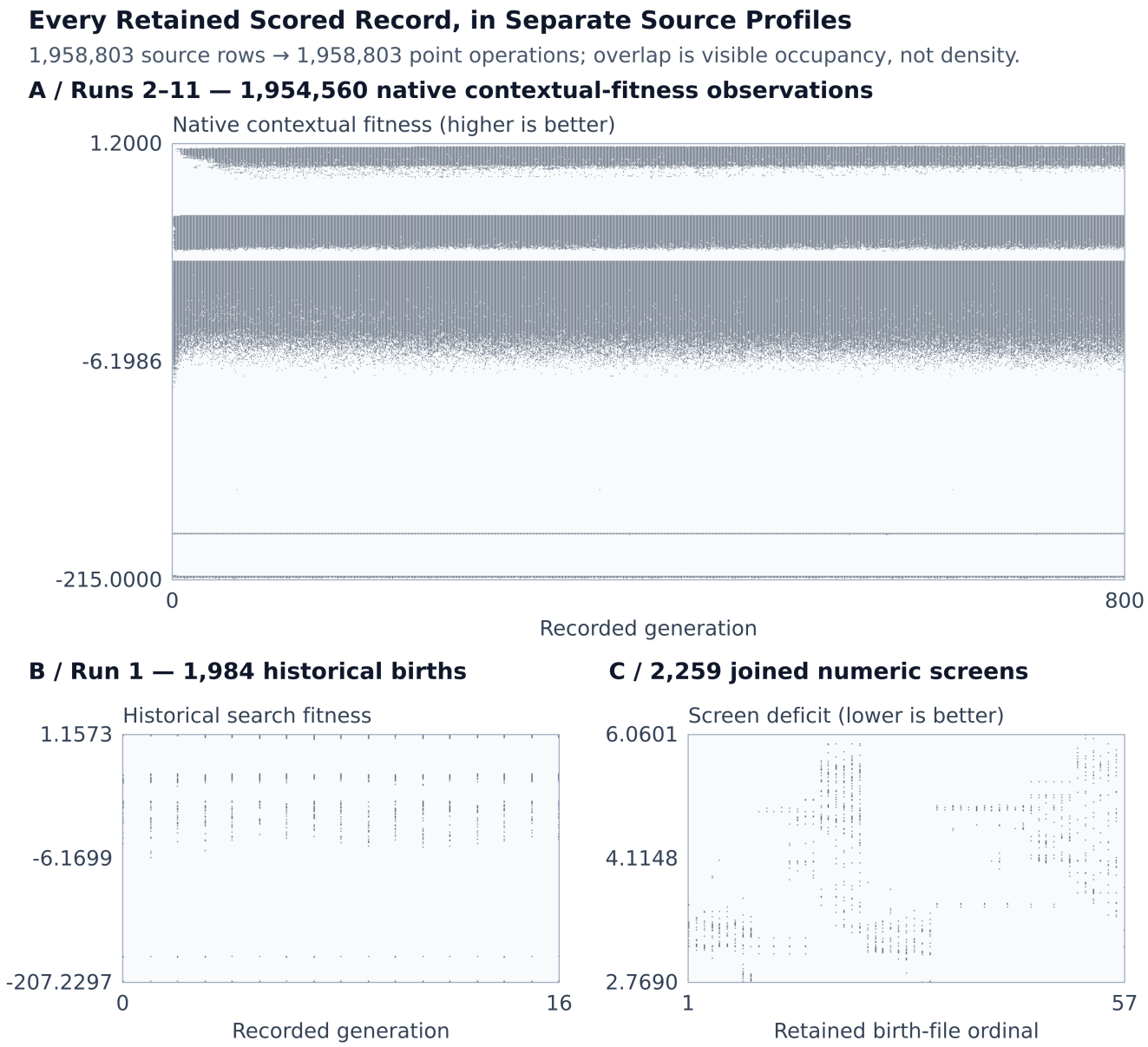


Fig. 5. Retained populations under separate score definitions: 1,954,560 native births, 1984 historical births and 2259 numeric screens. Each numeric record contributes a point; rasterized points can overlap. The separate nonnumeric band contains 1390 proposals without screen observations. The figure universe is these admitted records (1,958,803 numeric plus 1390 nonnumeric); zero excluded from this universe does not mean that every campaign, aborted attempt or historical record is included. Axes retain each regime's own score meaning.

IX. Numerical Agreement and Unresolved Error

A. Model, Load and Boundary Identity

The plate and local solid records solve linear-elastic static systems $\mathbf{Ku} = \mathbf{f}$ under their respective geometry, material, support and load assumptions. In the plate path, DKT bending supplies lateral response (MPa/N) and chord-torsion response (MPa/(N mm)); quadratic T6 membrane elements supply in-plane tip-pry and ring-traction response (MPa/N). Lateral and torsion values are per-triangle maxima over three corner-extrapolated bending stresses; pry and ring values are T6 centroid von Mises stresses. The envelope takes the maximum of lateral and pry responses scaled by $260\,\mathrm{N}$, ring traction scaled by $1500\,\mathrm{N}$, and torsion scaled by $1820\,\mathrm{N\,mm}$ with the producer's global torsion correction.

The local solid model uses quadratic (P_2) displacement fields on linear tetrahedral geometry. It clamps cut-arc vertices and edge-midpoint degrees of freedom to zero. Minimum-norm out-of-plane vertex forces on the $2\,\mathrm{mm}$ tip patch supply $1820\,\mathrm{N\,mm}$ chord torque, with zero resultant force and zero transverse moment. The retained producer does not transfer displacements from a larger model. This cut-arc boundary

TABLE VII
RETAINED OBSERVATION FAMILIES AND THEIR LIMITS. THE ENTRIES DESCRIBE AVAILABLE RECORDS, NOT NEWLY EXECUTED ANALYSES. FIELDS AND ASSESSMENT SCALARS SHOWN IN THE MAIN FIGURES BELONG TO INDIVIDUAL 2000.

Families	**Observable**	**Retained evidence**	**Supported interpretation**
Representation/export; reference shape	B-rep and mesh identity; mass (g); contour distance (mm) and similarity	Bound export records and reference comparison	Inspectable geometry and resemblance to an authored prior; no ergonomic approval
Lateral/pry; ring traction; plate torsion	Unit responses (MPa/N or MPa/(N mm)); capacities (N); regional factors	$0.25\,\mathrm{mm}^2$ cell fields; certification and strict scalars	Elastic responses under named loads and masks; no strict-mesh field is retained
Quadratic solid tip	Recovered vertex stress (MPa) and safety factors	Shared-mesh scalar pairs and distance-bin profiles	Agreement of selected recovered observables; no full solid stress array in the publication inputs
Support/stability; yield-order proxy	Supported regional ratio; provisional eigenvalue; elastic capacity ratio	Scalars and a three-level proxy sequence	Dependence on support/region assumptions; no physical fracture sequence
Fatigue	Modified-Goodman factor	Saved scalar $n_f = 0.472$	A model-based cyclic-load margin; no observed fatigue life
Discretization/solver consistency	GCI admission; relative difference of safety factors	Refused plate estimates; solid pairs differing by $< 0.05\%$	Separate evidence about mesh-study admission and shared-input agreement
Software/completeness controls	Contract outcomes, file completeness and score bookkeeping	Historical matrix and control definitions/receipts	Evidence organization and software behavior; no additional mechanical specimens

idealization is shared by both solid implementations.

B. Field Support and Composite Interpretation

For the 25,209 retained plate triangles, the envelope has 25,134 active cells and 75 producer-excluded cells within $4\,\mathrm{mm}$ of the boundary tip node. Its saved active peak is $1388.67\,\mathrm{MPa}$. Ring traction has 10,334 active cells and 14,875 producer-zeroed seated cells. The latter are masked; their zeros are not stress measurements. Lateral, tip-pry and raw torsion maps retain all 25,209 cells, including the tip region that is excluded from selected capacity calculations. Source-coordinate rounding makes one ring-boundary cell ambiguous geometrically; its positive stored ring response establishes that the producer retained it.

The five-map display summary averages active per-cell area-interval midpoint ranks, visits tied stored values in deterministic source-cell order, and then stretches the candidate composite from its own minimum/maximum to $[0, 1]$. It has five contributors on 10,334 triangles, four on 14,800 seated-region triangles and three on 75 tip-exclusion triangles. These counts describe the source support; the cover does not display a separate per-triangle contributor-count map. Its dimensionless values are relative to this candidate's retained fields. A value of 0.8 is not 80% safety or quality, and inclusion of the derived envelope does not create five independent mechanical confirmations. The Supplement records masks and the primitive-only comparison.

C. Plate Mesh Study: Refused Error Estimates

The retained three-level study uses maximum triangle areas 4, 1 and $0.25\,\mathrm{mm}^2$, corresponding to $h = \sqrt{A_{\max}} = 2$, 1 and $0.5\,\mathrm{mm}$ and 3166, 6803 and 25,209 elements. Its stored mesh/observable metadata are valid, but the selected relative-GCI procedure [12] admits neither error estimate:

- The yield-order proxy is monotone across the stored sequence but has observed order $p = -2.9003$; the procedure refuses a nonpositive order.
- The lateral safety-factor sequence has oscillatory differences and is refused on that basis.

The stored extrapolated values and fine-grid GCI estimates remain null. This is a failure to establish the assumptions of this retained procedure for these sequences, not a claim that every numerical-error method is impossible. The certification minimum over $0.25/0.125\,\mathrm{mm}^2$ and strict $0.0625\,\mathrm{mm}^2$ scalars are separate assessment stages; they do not supply this three-level study with an additional observation.

D. Shared-Mesh Solid Agreement

The primary solid path assembles scikit-fem vector P_2 tetrahedra [8] and uses PyAMG smoothed aggregation with CG. The comparison path assembles a DOLFINx 0.9.0 vector Lagrange P_2 space [9] and uses PETSc 3.22.0 CG/GAMG. Both consume the same saved mesh, elastic constants, cut clamp and patch forces. The scikit-fem/PyAMG executed package versions are not retained; the DOLFINx sidecar records its environment. The implementation table in the Supplement distinguishes these records from Rust orchestration.

Both recover von Mises stress at element vertices and form the incident-cell maximum or unweighted mean. The comparison region is $4 \le d_{\mathrm{tip}} < 15\,\mathrm{mm}$ with distance from the saved cut vertices $> 3\,\mathrm{mm}$. The saved safety factors are 0.669 versus 0.6693 under maximum recovery and 0.726 versus 0.7263 under mean recovery. With $\delta = |\mathrm{SF}_{\mathrm{DOLFINx}}/\mathrm{SF}_{\text{scikit-fem}} - 1|$, those rounded scalars give differences of 0.0448% and 0.0413%. The primary record

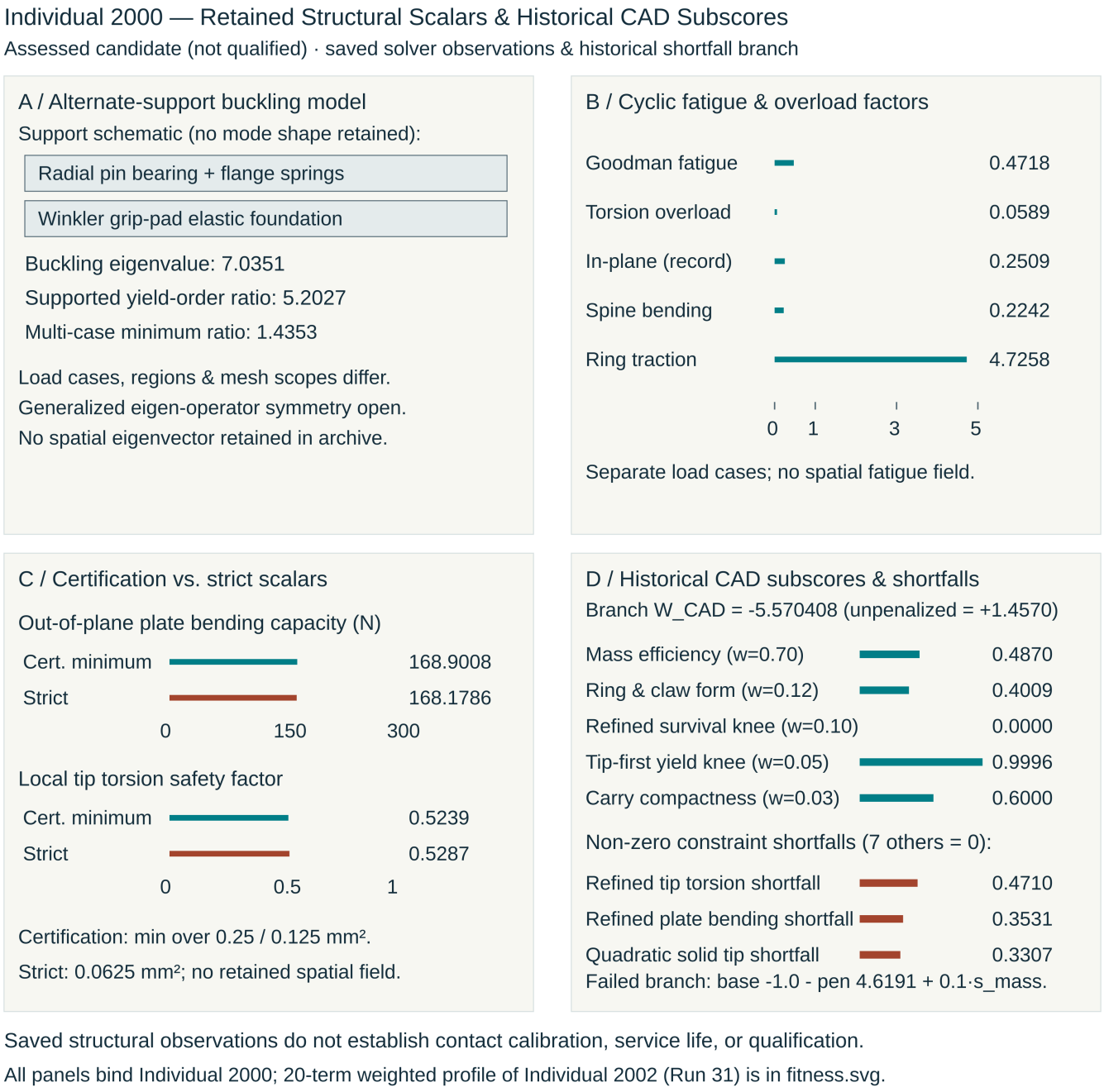


Fig. 6. Individual 2000: retained scalar assessment observations and conditional score bookkeeping. A: support assumptions and saved scalars; the eigen-operator audit remains open. B: modified-Goodman fatigue factor and human-overload response factors. C: certification minimum over $0.25/0.125\,\mathrm{mm}^2$ versus strict $0.0625\,\mathrm{mm}^2$ observations: lateral capacity 168.90 versus $168.18\,\mathrm{N}$ and local tip-torsion factor 0.524 versus 0.529. D: continuous CAD subscores and shortfalls for this assessed candidate, separate from the retained weighted reporting vector of Individual 2002. This is one assessed automatic export, with no qualified or promoted champion in its retained leaderboard.

reports 152 iterations and relative residual 8.80×10^{-11}; the comparison reports 158 iterations.

This agreement supports consistency of the selected recovered scalar observables under shared inputs. It does not verify every algebraic operation, bound discretization or modeling error, establish mesh independence, or validate the clamped submodel against physical measurements. The retained profile bins use different plate-cell and recovered-solid-vertex observables and therefore require their own interpretation.

E. Open Eigenproblem and Assurance Definitions

The support study's stored eigenvalue (7.035) is provisional. Its producer invokes a symmetric eigensolver on $\mathbf{K}_s^{-1}\mathbf{G}$. Symmetric component matrices alone do not establish Euclidean self-adjointness of that product, and no confirming operator-symmetry check or eigenpair residual is retained. The eigensolver's operator contract [11] identifies the unresolved question; it does not supply a historical validation result.

The frozen assurance catalog defines 178 challenges, or 356



Fig. 7. Individual 2000 retained numerical observations. A–B: plate sequences with maximum triangle areas $4/1/0.25\,\mathrm{mm}^2$; the stored relative-GCI estimates are refused for nonpositive order and oscillatory differences. C: scikit-fem/PyAMG and DOLFINx/PETSc safety factors on the same 91,008-tetrahedron, 416,583-degree-of-freedom solid submodel, using incident-cell vertex maximum and unweighted mean recovery. D: distance-bin stress maxima for raw plate cells and recovered solid vertex means; the differently recovered profiles are not a pointwise field-error estimate.

paired positive/negative control slots across 20 families. Those are definitions, not proof that all slots executed. Individual execution receipts and outcomes must establish which controls ran; unavailable receipts remain unavailable. Neither catalog size nor repeated software-control labels supplies additional candidate-specific mechanical evidence.

X. Reconstruction and Artifact Access

A. Four Reproduction Levels

B. Immutable Evidence Entry Points

The retained technical crosswalk provides the public-ID mappings and access to the historical matrix, export identities and original field inventory. The completed-population archive index identifies ten run manifests and 200 compressed segments. These links identify the prior retained evidence accession; they are not a claim that the present revised manuscript was authored at that revision. The final publication receipt supplies the matching manuscript/build accession and standalone supplement access. A TeX include fragment is source, not a compiled supplement download.

Candidate, producer, reporting-source and publication identities remain distinct. For example, a certificate hash binds content identity under its scheme; it does not prove a CAD artifact exists. The archived producer revision is preserved with its original inputs rather than replaced by the present manuscript revision. Hash equality establishes byte identity, not correctness of the original computation.

TABLE VIII
DISTINCT REPRODUCTION CLAIMS AND THEIR SCOPE IN THIS STUDY. THE PUBLICATION/RELEASE RECEIPT IDENTIFIES ACTUAL COMMAND OUTCOMES FOR ITS EXACT SOURCE AND OUTPUTS.

Level	Operation	Evidence boundary
Report reconstruction	Read retained contracts/archives; check identities; regenerate reporting views and documents	The publication operation; preserves evidence inputs but writes derived/build outputs
Solver replay	Rerun a bound CAD/FEM workflow under a declared environment	Not performed by the reporting rebuild
Software controls	Execute named positive/negative fixtures and admission checks	Actual receipts establish executed cases; catalog definitions alone do not
Physical replication	Fabricate specimens and obtain mechanical observations	Physical observations are not reported in this study

C. Reporting Commands and Writes

The reporting environment requires Rust/Cargo with the locked dependencies and complete admitted archives; document production additionally requires LuaLaTeX/IEEEtran, the declared fonts, librsvg and PDF inspection tools. From the repository root, the following existing entry points perform artifact auditing and document checks. Commands are shown at full width so flags remain copyable; actual exits/toolchain/output hashes belong to the final validation receipt.

The artifact audit decodes admitted historical frames, checks proposal/screen joins, verifies compressed-segment and reporting bindings, and checks generated-output consistency. The document check renders figures and compiles the paper under target/ieee/paper/, and the existing build also updates docs/paper/main.pdf. Evidence inputs remain read only; the entire operation is not filesystem read only. The supplement target, complete input manifest, all-page visual inspection and clickable-link checks are part of the matching publication package rather than implied by the existence of source files.

XI. PROSPECTIVE REPRESENTATION RESEARCH

FunSearch provides context for separating language-model program proposals from execution-based evaluation in mathematical and algorithmic settings [6]. It does not establish an agent result in Jarambit. Searching values in the present schema and proposing a different schema or decoder are distinct activities; the retained runs demonstrate the former.

A future controlled study on benign reference CAD tasks could compare human-authored and agent-assisted representation proposals under fixed evaluator versions and review criteria. A proposal would identify types, units, dependencies, serialization and migration rules, and expected tests; retained rejected proposals would permit measurement of invalidity, reconstruction success, and reviewer effort. Those outcomes remain unmeasured here. Human responsibility for the problem definition and interpretation remains essential, while independently controlled evaluators constrain which proposed effects are actually observed.

XII. DISCUSSION

A. What the Retained Links Add

The practical result is an inspectable distinction between a search preference and its supporting evidence. Parent links expose ancestor reuse; candidate and stage identities connect a stored ranking to a later assessment; source bindings keep selected numerical observations attached to their models. Missing observations remain visible rather than acquiring a favorable or unfavorable value. These capabilities explain the usefulness of the artifact without treating archive size or software size as proof of methodological novelty.

The numerical example makes the same point at another scale. Close agreement of selected observables on one shared solid discretization does not resolve the inadmissibility of a separate plate extrapolation. Both observations can be retained without collapsing them into one validation label. A derived view can help readers inspect such distinctions, but its source hashes alone do not establish the truth of the original computation.

B. Threats to Interpretation

The study is retrospective and confined to one repository. Prior workflows selected what was run and retained; completed runs are not a random sample of all attempts. Missing source files and later assessments restrict reconstruction, and missingness is not assumed random. Births share ancestry and search context, while numerical families can share geometry, fields, and assumptions. Record counts and family counts therefore do not count independent replications.

Contextual scores depend on the archive and population used to compute them. Cross-generation changes, per-run maxima, and cumulative best-so-far curves are descriptive observations rather than evidence of a common physical improvement trajectory. The historical assessment disagreement does not determine a population-wide failure rate. Open numerical questions, including the support eigenproblem audit, remain local limits on their associated observations.

Physical observations are not reported in this study. Material labels and CAD dimensions do not describe all relevant manufacturing state or measured process capability. NIST's test-artifact work illustrates why those properties require measurement [10]. Retained computational mass and response values consequently do not establish the performance of a manufactured part.

C. Questions Requiring New Evidence

A controlled benign case could test how often search rankings survive an independently specified later assessment. A reporting

```
rtk proxy cargo run --release --locked \
  --manifest-path crates/paper-artifacts/Cargo.toml \
  --example publication_v2_audit
rtk proxy make -C docs/paper check
```

Fig. 8. Reporting entry points, without invoking a new search, CAD export or mechanical solve. The artifact audit reads retained inputs and checks reporting consistency. The Makefile compiles figures/PDFs and writes build products.

study could measure whether independent readers reconstruct an observation correctly and detect provenance defects. Physical studies would need explicit specimens, conditions, metrology, and uncertainty. These prospective questions require new observations; the present contribution supplies a bounded evidence artifact from which to formulate them.

XIII. Conclusion

Jarambit links versioned candidate representations, retained search observations, and selected numerical assessments through explicit identities and source artifacts. The retrospective findings expose uneven evidence coverage, ancestor reuse and a later-assessment disagreement in the historical cohort, and distinct conclusions from shared-mesh solid agreement and refused plate extrapolation. This inspectable connection supports reconstruction and critical interpretation, while comparative search advantage, physical performance, and transfer to other design domains remain unestablished.

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References

[1] Jarambit project, "Publication source and retained-evidence snapshot," revision 102cdbef52b3710ce360ac34475c96ede0e33970, 2026. [Online]. Available: https://github.com/neverhuman/jarambit/tree/102cdbef52b3710ce360ac34475c96ede0e33970.

[2] R. Wu, C. Xiao, and C. Zheng, "DeepCAD: A deep generative network for computer-aided design models," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 6772–6782. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Wu_DeepCAD_A_Deep_Generative_Network_for_Computer-Aided_Design_Models_ICCV_2021_paper.html.

[3] M. S. Khan, S. Sinha, T. U. Sheikh, D. Stricker, S. A. Ali, and M. Z. Afzal, "Text2CAD: Generating sequential CAD designs from beginner-to-expert level text prompts," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 7552–7579, doi: 10.52202/079017-0242. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/0e5b96f97c1813bb75f6c28532c2ecc7-Abstract-Conference.html.

[4] J. D. Lohn, D. S. Linden, G. S. Hornby, W. F. Kraus, A. Rodríguez-Arroyo, and S. E. Seufert, "Evolutionary design of an X-band antenna for NASA's Space Technology 5 mission," NASA Technical Reports Server, document 20030067398, 2003, conference preprint. [Online]. Available: https://ntrs.nasa.gov/citations/20030067398.

[5] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," arXiv:1504.04909, 2015. [Online]. Available: https://arxiv.org/abs/1504.04909.

[6] B. Romera-Paredes *et al.*, "Mathematical discoveries from program search with large language models," *Nature*, vol. 625, pp. 468–475, 2024, doi: 10.1038/s41586-023-06924-6. [Online]. Available: https://doi.org/10.1038/s41586-023-06924-6.

[7] NASA, *Standard for Models and Simulations*, NASA-STD-7009B, Mar. 5, 2024. [Online]. Available: https://standards.nasa.gov/standard/NASA/NASA-STD-7009.

[8] T. Gustafsson and G. D. McBain, "scikit-fem: A Python package for finite element assembly," *J. Open Source Softw.*, vol. 5, no. 52, p. 2369, 2020, doi: 10.21105/joss.02369. [Online]. Available: https://joss.theoj.org/papers/10.21105/joss.02369.

[9] I. A. Baratta *et al.*, "DOLFINx: The next generation FEniCS problem solving environment," preprint, version 1, 2023, doi: 10.5281/zenodo.10447666. [Online]. Available: https://zenodo.org/records/10447666.

[10] S. Moylan, J. Slotwinski, A. Cooke, K. Jurrens, and M. A. Donmez, "An additive manufacturing test artifact," *J. Res. Natl. Inst. Stand. Technol.*, vol. 119, pp. 429–459, 2014, doi: 10.6028/jres.119.017. [Online]. Available: https://www.nist.gov/publications/additive-manufacturing-test-artifact.

[11] SciPy developers, "scipy.sparse.linalg.eigsh," API reference, accessed Oct. 20, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.linalg.eigsh.html.

[12] P. J. Roache, "Perspective: A method for uniform reporting of grid refinement studies," *J. Fluids Eng.*, vol. 116, no. 3, pp. 405–413, 1994, doi: 10.1115/1.2910291. [Online]. Available: https://doi.org/10.1115/1.2910291.

[13] O. Sigmund and K. Maute, "Topology optimization approaches: A comparative review," *Struct. Multidiscipl. Optim.*, vol. 48, no. 6, pp. 1031–1055, 2013, doi: 10.1007/s00158-013-0978-6.

[14] L. Moreau and P. Missier, Eds., *PROV-DM: The PROV Data Model*, W3C Recommendation, Apr. 30, 2013. [Online]. Available: https://www.w3.org/TR/2013/REC-prov-dm-20130430/.

[15] M. D. Wilkinson *et al.*, "The FAIR Guiding Principles for scientific data management and stewardship," *Sci. Data*, vol. 3, Art. no. 160018, 2016, doi: 10.1038/sdata.2016.18.

[16] A. M. Smith, D. S. Katz, K. E. Niemeyer, and FORCE11 Software Citation Working Group, "Software citation principles," *PeerJ Comput. Sci.*, vol. 2, e86, 2016, doi: 10.7717/peerj-cs.86.

[17] F. Crameri, G. E. Shephard, and P. J. Heron, "The misuse of colour in science communication," *Nat. Commun.*, vol. 11, Art. no. 5444, 2020, doi: 10.1038/s41467-020-19160-7.